Researching information about elements

Your teacher will give you the name of an element. You probably know nothing about it.

Your task is to do some research using a data book, a CD-ROM or an internet site.

Use the list here to help you.

Name of element	Symbol
Date of discovery	Density
Boiling point	Melting point
Appearance	Uses
Position in the Periodic Table	Other information you find out

Find out what you can about the element and summarise all your information on a piece of paper.

When the whole class has finished, you can look at all the pieces of paper. See if you can find any elements with similar properties.

The class could produce a display of the information everyone collected.



Extension

E3

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New compounds

Look at the Periodic Table.

							H hydrogen										He helium
Li lithium	Be beryllium											B boron	C carbon	N nitrogen	O oxygen	F fluorine	Ne neon
Na sodium	Mg magnesium											Al aluminium	Si silicon	P phosphorus	S sulphur	Cl chlorine	Ar argon
K potassium	Ca calcium	Sc scandium	Ti titanium	V vanadium	Cr chromium	Mn manganese	Fe iron	Co cobalt	Ni nickel	Cu copper	Zn zinc	Ga gallium	Ge germanium	As arsenic	Se selenium	Br bromine	Kr krypton
Rb rubidium	Sr strontium	Y yttrium	Zr zirconium	Nb niobium	Mo molybdenum	Tc technetium	Ru ruthenium	Rh rhodium	Pd palladium	Ag silver	Cd cadmium	In indium	Sn tin	Sb antimony	Te tellurium	I iodine	Xe xenon
Cs caesium	Ba barium	La lanthanum	Hf hafnium	Ta tantalum	W tungsten	Re rhenium	Os osmium	Ir iridium	Pt platinum	Au gold	Hg mercury	Tl thallium	Pb lead	Bi bismuth	Po polonium	At astatine	Rn radon
Fr francium	Ra radium	Ac actinium															

Find the column of elements on the far right-hand side. They are called the **noble gases**. These are helium, neon, argon, krypton, xenon and radon.

Until about fifty years ago, scientists thought these gases did not react at all under any conditions.

In 1962 Neil Bartlett made the first compound of a noble gas. It was a yellow-orange solid. Diagram A shows a simplified view of the atoms in this compound.

This compound contains molecules made up of atoms of xenon, platinum and fluorine.

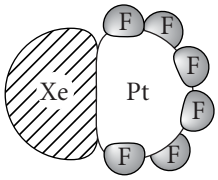
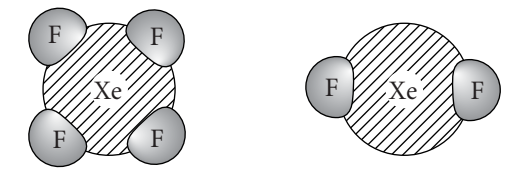


diagram A

a How many atoms of each element are there in a molecule of this compound?

b What is the total number of atoms in one molecule of this compound?

The discovery of this compound made scientists look for others. They soon found xenon tetrafluoride and xenon difluoride, shown in diagram B.



xenon tetrafluoride xenon difluoride

diagram B

c Write down how many atoms there are of each type in xenon tetrafluoride and xenon difluoride.

d Describe the shape of xenon tetrafluoride.

Finding these compounds of xenon changed scientists' ideas about noble gases and they went on to look for other compounds of noble gases. There were still three noble gases that they thought did not form any compounds – helium, neon and argon. Sometimes the compounds they tried to make were very unstable and split up again.

In 2000, scientists in Finland made a compound of argon. This is shown in diagram C.

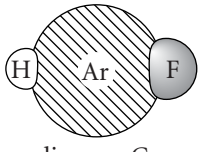


diagram C

e Write down the three elements in this compound of argon.

f How many atoms are there in a molecule of this compound?

Having succeeded in making this compound, scientists will continue to try to make compounds of the other noble gases.

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Combining power

The **combining power** of an element is the number of hydrogen atoms that will combine with one atom of the element.

Here are the combining powers of different elements.

Combining power of 1

- sodium
- potassium
- fluorine
- chlorine
- bromine
- iodine
- hydrogen

Combining power of 2

- oxygen
- sulphur
- magnesium
- calcium
- barium
- zinc

Combining power of 3

- nitrogen
- aluminium

Combining power of 4

- carbon

These combining powers can be used to write the formula for a compound.
Let's look at some ways we can use combining power.

Example 1: carbon and chlorine

Carbon has a combining power of 4 and chlorine has a combining power of 1. Each carbon will combine with 4 chlorines.

The formula is CCl_4.

Example 2: aluminium and sulphur

Aluminium has a combining power of 3 and sulphur has a combining power of 2.

So the formula of aluminium sulphide is Al_2S_3. Two aluminium atoms provide 6 links and three sulphurs provide 6 links.

Now use combining powers to write the formula for each of the following compounds.

1 sodium oxide
2 sodium iodide
3 potassium fluoride
4 potassium oxide
5 magnesium chloride
6 magnesium nitride
7 calcium chloride
8 aluminium carbide
9 hydrogen sulphide
10 hydrogen chloride

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Other types of mixture

A sugar solution is a mixture of sugar and water. The sugar particles are spread evenly throughout the solution. The solution looks just like water. It is completely transparent. You only know the sugar is there if you taste it.

There are other types of mixture. Many of these mixtures have great importance in everyday life.

A **colloid** is made of two substances which mix thoroughly but do not dissolve in each other. There are different types of colloid.

One type of colloid is an **emulsion**. Here one substance, in the form of tiny particles, is **dispersed** in (spread through) the other.

Milk is an emulsion with tiny droplets of fat dispersed in a watery layer. The fat droplets are very small and do not settle out. Milk is called an **oil-in-water emulsion**.

1 Butter is a water-in-oil emulsion. Suggest how this is different from milk.

2 Other mixtures include:

shaving cream	**salad cream**	**black coffee**	**wallpaper paste**

Copy and complete the table using mixtures from the list.

Type of mixture	**The mixture is**	**Example**
solution	one substance dissolved in another	
foam	a gas mixed with a liquid	
emulsion	two liquids that do not dissolve in each other but are dispersed through each other to give a thick liquid that pours slowly	
gel	a liquid dispersed in a solid to give a jelly-like consistency	

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Why do rocks expand when heated?

The Kinetic Theory

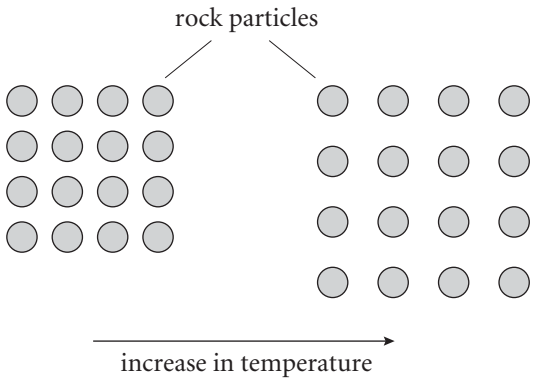
The movement of particles in solids, liquids and gases is described by the Kinetic Theory. Particles in a solid are fixed in position and cannot move from place to place. They are, however, able to vibrate.

The particles in a solid lump of rock move from side to side, but around a fixed position. This vibration needs energy. In theory, if a particle has no energy it will no longer vibrate. This will happen at the temperature called **absolute zero**, which is −273 °C. No one has managed to get anything this cold yet!

As the temperature of the rock rises, the particles gain more energy. This means that they vibrate more. The higher the temperature, the more the particles vibrate. Eventually a temperature is reached where the particles vibrate so much that they break apart from each other, and can move around. This temperature is the melting point of the rock.

When solid rock is heated up from one temperature to another, but not as high as its melting point, the rock expands. This means that its volume increases, but its mass stays the same. The rock has exactly the same number of particles at the higher temperature, but they are spread out more. Why does this happen?

As the particles gain energy and vibrate more, they need more room for this movement. The particles move a little further apart. The higher the temperature the further they move apart. This is the reason why the rock expands.



Note that the particles themselves do not get any bigger, they simply take up more space. It is like a group of people standing very close together watching a football match. As long as they do not move much they can stand very close. If they start to jump around they need more space. This happens when a goal is scored, and can cause a surge in the crowd. People standing at the front who have nowhere to move to can get squashed. Modern football stadiums have spectators sitting down to prevent this.

Question

A pen friend gets poor marks in a test about the expansion of rocks. He writes to say that his marks were low because he cannot understand how a solid lump of rock can get bigger when it gets hotter if the particles of the rock do not get bigger.

Write a letter to help improve the pen friend's marks when he re-takes the test.

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